Indicator and Protection for a Laboratory Power Supply

STM32F103C8T6 + INA226 + ST7735

A high-accuracy, low-cost volt/ammeter with hardware overload monitoring, manual ALERT reset and software I2C that works around the lock-up of the STM32F103 hardware I2C peripheral.

Platform	Blue Pill, STM32F103C8T6
Meter	INA226, 0.033 Ω shunt
Interface	ST7735, rotary encoder, buzzer
Protection	SOL + LEN, active-LOW ALERT

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English technical description, pinout and commissioning procedure

GitHub: Aleksei327 / STM32F103-INA226-ST7735

Video: youtu.be/KYVpQovXies

Contents

Document sections and subsections. Page numbers are updated automatically when the PDF is built.

What is implemented in the local code 3

- What the 0x484F configuration means 3

Overall Blue Pill pin table 4

ST7735 display connections 4

Encoder connections 4

INA226 connections 5

- Pins of the INA226 chip itself 5
- Pins of a typical INA226 module 5

How ALERT and the reset button work 5

- What “resetting ALERT by reading register 0x06” means 6

Buzzer connection 6

Commissioning procedure 6

Differences between the local code and the description on GitHub 6

Main project files 7

This design lets you build a high-accuracy digital volt/ammeter and a protection system for a laboratory power supply at a very low cost. The device uses readily available modules: STM32F103C8T6 (Blue Pill), INA226, an ST7735 TFT display, a mechanical rotary encoder and a passive buzzer.

On the hardware this project was created for, the STM32F103 hardware I2C peripheral hung when working with the INA226, which made the instrument unreliable. This problem was solved in software: all communication with the INA226 is handled by a custom software I2C implementation (bit-banging) on pins PB8 and PB9, without using the microcontroller's hardware I2C controller.

Only the I2C transfers are done in software. Voltage and current measurement, comparison of the shunt voltage against the configured threshold, and generation of the ALERT signal are performed in hardware inside the INA226. As a result, the protection does not depend on the speed of the STM32 main loop.

Links:

- GitHub: Aleksei327/STM32F103-INA226-ST7735-Indicator-Encoder-Protection-PS
- YouTube video: project demonstration
- INA226 documentation: Texas Instruments INA226

What is implemented in the local code

- display of voltage, current, power, accumulated charge and the configured current limit on the ST7735 display;
- communication with the display over hardware SPI1;
- communication with the INA226 over software I2C on GPIO;
- INA226 presence check at start-up;
- measurement of voltage, current and power;
- setting the protection threshold by turning the encoder;
- configuration of the INA226 ALERT output for shunt voltage over-limit (SOL, Shunt Voltage Over-Limit);
- active-low ALERT output;
- LEN latch mode in the INA226;
- confirmed on the assembled device: the protection and the physical ALERT output are held in the active state until a manual reset;
- an ALERT! screen and an intermittent audible signal when a fault is detected;
- reset of the software fault state with the encoder button;
- a 0.5-second buzzer test at power-up.

Current firmware parameters:

Parameter	Value
Initial current limit	0.15 A
Minimum limit	0.01 A
Maximum limit	2.35 A
Normal encoder step	0.01 A
Step during fast rotation	0.1 A
Shunt resistance	0.033 Ω
Current measurement resolution	0.001 A
INA226 address, 7-bit	0x40
Address in firmware format, with the R/W bit	0x80
INA226 configuration	0x484F
Software voltage correction	+0.015 V
Encoder debounce	15 ms
Buzzer frequency	approximately 2.5 kHz

What the 0x484F configuration means

0x484F is a 16-bit value of the INA226 Configuration register. It is not a program version number or an error code. In the current firmware it sets the following parameters:

INA226 field	Value
AVG averaging	128 samples
Bus voltage conversion time VBUSCT	204 µs
Shunt voltage conversion time VSHCT	204 µs
MODE	continuous shunt and bus voltage measurement

One complete averaged shunt-and-bus measurement cycle takes approximately:

128 × (204 µs + 204 µs) = 52.2 ms.

The values in the table on GitHub are not newer versions — they are simply other valid settings of the same register:

Value	Averaging	Time for one full cycle
0x4127	1 sample	about 2.2 ms
0x4527	16 samples	about 35.2 ms
0x4927	128 samples	about 281.6 ms
0x484F, currently used	128 samples	about 52.2 ms

In other words, 0x484F keeps the heavy 128-sample averaging of 0x4927 but uses a shorter conversion time for each sample. This is a compromise between steady readings and reaction speed.

Overall Blue Pill pin table

Blue Pill	Function	Connected pin
PA0	EXTI0 input	encoder CLK
PA1	digital input	encoder DT
PA2	reset button, active LOW	encoder SW or a separate button
PA3	TIM2_CH4, PWM ≈ 2.5 kHz	passive buzzer via a suitable driver
PA5	SPI1_SCK	ST7735 display SCL, SCK or CLK
PA7	SPI1_MOSI	ST7735 display SDA, MOSI or DIN
PB0	digital output	ST7735 display RES or RST
PB1	digital output	ST7735 display DC, A0 or RS
PB8	software I2C SCL	INA226 SCL
PB9	software I2C SDA	INA226 SDA
PB10	digital output	ST7735 display CS or SS
3.3V	supply	logic of ST7735, INA226 and the encoder
GND	common ground	all modules and the external protection circuit

Pins PB6 and PB7, listed in the CubeMX file as hardware I2C1, are not used in the working `main()` function. The INA226 is served by software I2C on PB8 and PB9.

ST7735 display connections

Pin names differ from module to module. Use the function and the alternative designations in the table for orientation.

ST7735 pin	Connect to Blue Pill	Function
VCC	3.3V	display supply
GND	GND	common ground
SCL, SCK, CLK	PA5	SPI1 clock line
SDA, MOSI, DIN	PA7	data from the STM32 to the display
RES, RST, RESET	PB0	display hardware reset
DC, A0, RS	PB1	command/data select
CS, SS	PB10	display select
LED, BL, BLK	3.3V	backlight
MISO, SDO	not connected	the firmware does not read data from the display

It is safer to power the module and its logic inputs from 3.3 V. If a particular module has no current-limiting resistor for the backlight LED, a resistor must be added in series with the LED/BL pin.

The code is set up for a 160×128 display with an ST7735S controller in “rotate left” orientation. The settings are in `Core/Inc/st7735.h`.

Encoder connections

For the widely used KY-040 module or a compatible one:

Encoder pin	Connect to Blue Pill	Function
+, VCC	3.3V	module supply
GND	GND	common ground
CLK	PA0	encoder pulses, EXTI0 interrupt

DT	PA1	direction detection
SW	PA2	fault-state reset button

Inputs PA0, PA1 and PA2 have internal pull-ups to 3.3 V. When pressed, the button must short PA2 to GND. For long cables, external 4.7–10 kΩ pull-up resistors from CLK, DT and SW to 3.3 V may be fitted.

If the adjustment direction turns out to be reversed, swap CLK and DT, or change the counter direction in `EXTI0_IRQHandler()` in `Core/Src/encoder.c`.

INA226 connections

Pins of the INA226 chip itself

The table corresponds to the DGS package, VSSOP-10. A ready-made module may have a different pin order, so check the markings on the board.

No.	INA226 pin	Connection
1	A1	GND, for address 0x40
2	A0	GND, for address 0x40
3	ALERT	10 kΩ pull-up to 3.3 V and input of the external shutdown circuit
4	SDA	Blue Pill PB9
5	SCL	Blue Pill PB8
6	VS	Blue Pill 3.3V
7	GND	common GND
8	VBUS	measured voltage, normally the load side after the shunt
9	IN−	shunt side going to the load
10	IN+	shunt side going to the positive supply rail

VS is the chip supply and accepts only 2.7–5.5 V. It must not be connected to the power rail of the laboratory power supply if that rail exceeds 5.5 V. The VBUS input is intended for measuring the power rail and, on the INA226, accepts up to 36 V.

Pins of a typical INA226 module

Module pin	Connection
VCC, VS	Blue Pill 3.3V
GND	common GND
SCL	Blue Pill PB8
SDA	Blue Pill PB9
A0	GND
A1	GND
ALERT, AL	10 kΩ pull-up to 3.3 V and the external shutdown circuit
IN+, VIN+	positive supply, before the shunt
IN−, VIN−	after the shunt, to the positive side of the load
VBUS	to IN−, if this pin is not already tied to it on the module

On many modules A0, A1, VBUS and the shunt are already connected by PCB traces. Before wiring, check the schematic of your particular module with a multimeter or against the seller's documentation.

The SCL and SDA lines are configured as open-drain without internal pull-ups. External pull-up resistors to 3.3 V are required. The code assumes 10 kΩ; most ready-made INA226 modules already have pull-ups fitted.

How ALERT and the reset button work

1. The firmware calculates the voltage drop across the shunt:
   `V_threshold = I_threshold × 0.033 Ω`
2. The resulting voltage is converted into an Alert Limit register value with a step of 2.5 µV.
3. The following are enabled in the Mask/Enable register:
   - SOL — shunt voltage over-limit;
   - LEN — latch mode;
   - APOL = 0 — active-low level.
4. The INA226 compares the shunt measurement result against the threshold in hardware. On overcurrent it sets the AFF flag and drives the physical ALERT output LOW.
5. As long as no software fault has been registered yet, the STM32 reads the Mask/Enable register (0x06) and checks the AFF flag.

6. Having detected AFF, the STM32 sets its own `alert_triggered` flag. After that, ALERT! blinks on the display, the buzzer sounds intermittently, and the check function stops re-reading register 0x06.
7. On the assembled device, the protection and the physical INA226 ALERT output have been confirmed to stay active until a manual reset.
8. When the encoder SW is pressed, input PA2 is shorted to GND. The STM32 calls `INA226_ClearAlert()`, clears its `alert_triggered` flag and re-enables SOL + LEN mode in the INA226.
9. If the current is already below the threshold, the device continues normal operation. If the overload persists, the INA226 will set AFF again at the next measurement, activate ALERT, and the program will return to the fault state.

ALERT is an open-drain output. A pull-up of approximately 10 kΩ to 3.3 V is mandatory. This pin is intended for a logic-level signal and must not be connected directly to the gate of a power MOSFET or to high voltage. A suitable shutdown circuit is required between ALERT and the power stage: a transistor driver, an optocoupler, a comparator and/or a hardware latch.

In the current firmware, ALERT is not connected to a separate Blue Pill GPIO. The STM32 learns about the fault by reading the AFF flag over I2C. The ALERT output itself must go to an external hardware shutdown circuit. Its exact schematic is not present in the repository, so it is not possible to assign specific pins of an external transistor, comparator or power MOSFET from this code alone.

What “resetting ALERT by reading register 0x06” means

The INA226 has no dedicated I2C command called “clear ALERT”. According to the Texas Instruments documentation, with LEN = 1 the latch is acknowledged and released by reading the Mask/Enable register (0x06).

In other words, the statement “the STM32 clears ALERT in software” means the following:

1. The STM32 receives a button press on PA2.
2. The `INA226_ClearAlert()` function reads register 0x06 over software I2C.
3. The INA226 treats this read as an acknowledgement of the fault and releases its internal latch.
4. The STM32 clears the software fault state and writes the SOL + LEN settings again.
5. The INA226 continues to monitor the current in hardware.
6. At normal current, ALERT stays released; if the overload persists, the INA226 activates ALERT again at the next conversion.

Ordinary reads of the voltage, current and power registers do not clear the latch. This behaviour is tied specifically to reading the Mask/Enable register (0x06).

In the current program, register 0x06 is also read once in `INA226_IsAlertTriggered()` so that the STM32 can see the AFF flag. Despite this, on the assembled device the physical ALERT output and the protection have been verified to hold until the button is pressed. Functionally, the device works exactly as described above: the button issues the software reset command, after which the INA226 re-checks the current and either resumes operation or re-engages the protection.

Buzzer connection

PA3 generates PWM at approximately 2.5 kHz, so the firmware is designed for a passive buzzer.

For a low-power piezo element the following is acceptable:

Pin	Connection
+	PA3
−	GND

If the buzzer draws more current than the GPIO permits, it must be driven through an NPN transistor or a MOSFET with a common GND. A high-power or electromagnetic buzzer must not be powered directly from PA3.

Commissioning procedure

1. Make all connections only with the power off.
2. Check the common ground of the Blue Pill, ST7735, INA226, encoder and protection circuit.
3. Make sure the INA226 VS receives 3.3 V and not the power rail voltage.
4. Check the pull-ups on SCL, SDA and ALERT to 3.3 V.
5. At power-up the buzzer should sound for 0.5 seconds.
6. The display should show `Init INA226...` and then `INA226 OK!`.
7. Check that the limit can be changed with the encoder, with no heavy load connected.
8. Test the hardware shutdown first from a laboratory source with a low current limit and with a safe test load.
9. Separately check the ALERT level with an oscilloscope, and that the shutdown is held after the overload has been removed.

Differences between the local code and the description on GitHub

- In the code the maximum is 2.35 A, not 2.38 A.
- The initial limit is 0.15 A, although the comment next to the variable says 1.00 A.
- The voltage correction is +0.015 V, not +0.020 V.
- The working code uses 0x484F: 128-sample averaging and a short conversion time of 204 µs. The values on GitHub are alternative speed and averaging settings, not different firmware versions.
- The display is redrawn on every cycle; updating only the values that have changed is not implemented.
- There is no separate GPIO for reading the physical ALERT output.

- On the assembled device, the software reset of the INA226, the re-check of the current and the holding of the physical ALERT output until the button is pressed have been confirmed in practice.
- During the audit of `Core/Src/main.c`, a typo was found and fixed in the font name on the fault screen: `Font_11x1e8` (with a Cyrillic “e”) was replaced with `Font_11x18`.

Main project files

- `Core/Src/main.c` — main loop, user interface and fault state;
- `Core/Src/INA226.c` and `Core/Inc/INA226.h` — measurements and ALERT configuration;
- `Core/Src/soft_i2c.c` — software I2C on PB8/PB9;
- `Core/Src/encoder.c` — encoder, button and buzzer;
- `Core/Src/st7735.c` and `Core/Inc/st7735.h` — display driver;
- `Core/Src/spi.c` — SPI1 configuration.